

UniAdapFuse (AdapFuse-UAV) — Project Summary

Status as of 21 September 2026 | Sources: PROJECT_EVALUATION.md, EXPERIMENT_RUNBOOK.md, abstract, results CSV

1. What we did

UAV disaster perception thesis with two tracks:

- **Scene classification (AdapFuse-v1):** fuses RGB + thermal + audio to predict disaster class and victim presence; works with missing sensors.
- **Smoke/fire detection** on D-Fire with YOLO models, plus our **Hazard Prior Gate (HPG)** injecting flame-colour, smoke and edge cues into the detector.
- Evidence audit of the thesis, figures rebuilt from real logs only, grouped (leakage-safe) split and repeated-seed tooling added.

2. Models

- **AdapFuse-v1:** encoders MobileNetV3-S (RGB), ResNet18 (thermal), small CNN (audio) → 192-d tokens → RUE sigmoid gates → 3-token cross-modal attention → GRU (sequence length 1, not truly temporal) → disaster, victim and (unsupervised) nuisance heads.
- **Comparisons:** single-modality baselines A1–A3; fusion designs B–F (early, late, intermediate, v1, v2 with knowledge distillation); UniAdapFuse-v2 joint classification + detection prototype.
- **Detectors:** YOLOv8n, v9t, v10n, v11n, YOLO26n/s, RT-DETR-R18, EfficientDet-D0, Faster R-CNN and SSD (MobileNetV3), HPG-YOLO26n.

3. Data flow

- **Sources:** FLAME, AIDER, C2A, SARD, ESC-50 → one metadata table of 72,997 rows (51,097 train / 10,949 val / 10,951 test).
- **Inputs:** RGB image; pseudo-thermal (derived from RGB, no real IR sensor); ESC-50 audio paired by label (2,000 clips reused).
- **Classification pipeline:** audio → log-mel spectrogram; all three → encoders → gates → attention → GRU → heads.
- **Detection pipeline:** D-Fire images → CLAHE enhancement → YOLO / HPG-YOLO → smoke and fire boxes.

4. Results

Model	Result	Note
AdapFuse-v1	98.74% combined macro-F1	Row-level split
RGB-only baseline	98.85%	Fusion gives no gain
Design D	98.69%	
AdapFuse-v1, grouped split	89.14%	Reveals leakage in row-level split
UniAdapFuse-v2	98.57% F1; detection mAP50 = 0	Joint detection not working
Phased training	Recovers F1 from 53.71% collapse	Stop-gradient isolation
YOLO26s + CLAHE (D-Fire)	0.771 mAP50 / 0.448 mAP50:95	Best result
YOLO26n baseline	0.7147 mAP50, 4.45 h	
HPG-YOLO26n	0.7053 mAP50, 1.96 h	2.27× faster training
Latency	14.26 ms / 70 FPS	RTX 5090, not embedded

Known weak points:

- No real thermal camera data and no synchronised audio.
- RUE gates nearly constant (RGB/thermal ≈ 0.97 , audio ≈ 0.006); they do not react to corruption.
- Nuisance and degradation losses are zero: trainer passes None for their labels.

5. Paths

What	Path
Classification code	code_p2/adaptfuse_uav/ (models/full_model.py, training/trainer.py, scripts/build_grouped_splits.py, configs/adaptfuse_v1_grouped_gtx960.yaml)
Grouped split	code_p2/adaptfuse_uav/data/metadata_grouped/seed_42/
Detection code	adaptfuse_od/ (training/run_efficient_study.py, models/novel_modules/)
Detection results	adaptfuse_od/outputs/experiment_results_table.csv, outputs/accuracy_study/, outputs/efficient_study/
Thesis	main.tex, chapters/, core/abstract.tex, images/generated/
Docs	PROJECT_EVALUATION.md, EXPERIMENT_RUNBOOK.md, ADAPTFUSE_UAV_COMPLETE_REFERENCE.md

6. Next steps (runbook order)

- 1 Grouped split with 5 seeds (GTX 960: about 1–3 days per seed).
- 2 SARD person bounding-box evaluation.
- 3 Fix the joint detector until mAP is above zero.
- 4 Add nuisance / degradation labels.
- 5 Feed real frame sequences to the GRU.
- 6 Add registered IR and synchronised UAV audio.
- 7 Benchmark on embedded UAV hardware.